12 (a) Find the value of $81^{-\frac{1}{2}}$

Negative power: take the reciprocal.

$81^{\frac{-1}{2}} = \frac{1}{\sqrt{81}} = \frac{1}{9}$

1 mark
Use reciprocal or square root

1 mark
Correct value

$\frac{1}{9}$

(2)

(b) Find the value of $\left(\frac{64}{125}\right)^{\frac{2}{3}}$

Cube roots: 64 → 4 and 125 → 5.

$\frac{4}{5} \times \frac{4}{5} = \frac{16}{25}$

1 mark
Cube root, then square

1 mark
Correct value

$\frac{16}{25}$

(2)

(Total for Question 12 is 4 marks)

13 The table shows a set of values for x and y.

x	1	2	3	4
y	9	$2\frac{1}{4}$	1	$\frac{9}{16}$

y is inversely proportional to the square of x.

(a) Find an equation for y in terms of x.

.....

(2)

(b) Find the positive value of x when $y = 16$

.....

(2)

(Total for Question 13 is 4 marks)

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